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16 JULY 1958
SUPERSEDING
MIL-I-16182A
24 JANUARY 1952

MILITARY SPECIFICATION

BRICKS, INSULATING, HIGH TEMPERATURE, DIATOMACEOUS SILICA

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers high-temperature insulation brick for use as a back-up insulation for refractory furnace linings of Naval boiler furnaces.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-S-781 — Strapping, Flat; Steel.

PPP-B-601 — Boxes, Wood, Cleated-Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Boxes, Fiber.

MILITARY

JAN-P-108 — Packaging and Packing for Overseas Shipment — Boxes, Fiberboard (V-Board and W-Board), Exterior and interior.

MIL-L-10547 — Liners, Case, Waterproof.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

FSC 9350

FOR FICHE

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OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Material. The brick shall be composed of diatomaceous silica which has been burned or fined to produce the required strength, low heat conductivity, and structure.

3.2 Dimensions.

3.2.1 Unless otherwise specified by the bureau or agency concerned the size of the brick shall be either 9 by 4½ by 2½ inches or 9 by 4½ by 1¼ inches as specified (see 6.1).

3.2.2 *Tolerances.*

3.2.2.1 Standard 9 by 4½ by 2½ inch straight brick shall not vary from the specified dimensions by more than 1/16 inch in width or thickness and 3/32 inch in length.

3.2.2.2 The 9 by 4½ by 1¼ inch split brick shall not vary from the specified dimensions more than that permitted for the standard size brick (see 3.2.2.1), except that the thickness shall not vary more than 1/32 inch.

3.2.2.3 For special shapes, no dimensions shall vary more than 2 percent from the dimensions specified, except that the maximum variation permitted will be 1/16 inch.

3.3 Density. The average weight shall be not more than 2.50 pounds for the 9 by 4½ by 2½ inch straight brick, and not more than 1.30 pounds for the 9 by 4½ by 1¼ inch split brick, approximately 43 and 45 pounds, respectively, per cubic foot (see 4.3.1).

3.4 Pyrometric cone equivalent. The pyrometric cone equivalent shall average not less than cone 20 (2,786°F.) (see 4.3.2).

3.5 Modulus of rupture. The modulus of rupture shall average not less than 65 pounds per square inch (p.s.i.) and not more than two of the 10 bricks tested shall show less than 45 p.s.i. (see 4.3.3).

3.6 Thermal conductivity. The thermal conductivity expressed in British thermal units (B.t.u.) per square foot per hour per degrees F. per 1 inch thickness shall be as follows (see 4.3.4).

Mean temperature (°F.)	*Thermal conductivity* (k.) (max.)
500	1.7
1,000	1.9
2,000	2.3

3.7 Simulative service. Bricks shall pass the simulative service test specified in 4.3.5.

3.8 Linear change. Bricks shall show an average linear shrinkage of not more than 1 percent or a linear expansion of not more than 2 percent on the hot face when tested as specified in 4.3.5 and 4.3.6.

3.9 Marking. Bricks shall be marked with the manufacturer's name or trade-mark by suitable indentation or stamping.

3.10 Workmanship. Bricks shall be of homogeneous structure, free from cracks, laminations, segregations, void defects, or soft centers. All corners and edges shall be sufficiently strong to prevent excessive crumbling or chipping when handled or shipped.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling.

4.1.1 *Lot for purposes of examination.* All bricks of the same shape and size, not exceeding 10,000 presented at one time, shall be considered a lot for purposes of acceptance examination.

4.1.2 *Lot for purposes of lot acceptance tests.* All bricks not exceeding 10,000, pre-

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sented at one time without restrictions as to shape and size, shall be considered a lot for purposes of tests.

4.1.3 *Sampling for examination of packaging.* A random sample of packages filled with bricks shall be selected for Government inspection in accordance with Standard MIL–STD–105 at Inspection Level I. The Acceptable Quality Level shall be equal to 2.5 percent defective to verify compliance with section 5.

4.1.4 *Sampling for examination of brick.* A random sample of brick shall be selected for Government inspection in accordance with Standard MIL–STD–105 at Inspection Level II. The Acceptable Quality Level shall be equal to 2.5 percent defective. The bricks shall be selected from packages selected in the packaging inspection sample.

4.1.5 *Sampling for lot acceptance tests at place of manufacture.* From each lot the Government inspector shall select a sufficient number of representative sample bricks in accordance with table I for the tests specified in 4.2.4.

TABLE I. *Sampling for tests.*

Lot acceptance test	Applicable paragraph	Number of bricks in sample
Density	4.2.4.3	10 for test
Pyrometric cone equivalent	4.2.4.4	5 for test and 5 for retest
Modulus of rupture	4.2.4.5	10 for test

4.1.6 *Sampling procedure for tests at a Government laboratory.* From the first lot of brick offered for delivery under a contract or order, and thereafter from one lot in each group of 10 successive lots, the Government inspector shall select 25 representative bricks of standard shape from different portions of the lot. Moreover, when the lot in question contains bricks of other than standard shape, three additional bricks of each such nonstandard shape shall be included. Samples shall be forwarded to the Naval Boiler and Turbine Laboratory, Naval Base Station, Philadelphia 12, Pa. Each sample brick shall be properly marked with the contractor's (manufacturer's) name, brand name, class, contract or order number, car number and destination.

4.2 Inspection.

4.2.1 *Examination of packaging.* Each of the sample packages selected in accordance with 4.2.3, shall be examined for defects of construction, for unsatisfactory markings, and for noncompliance with the requirements specified in section 5. If the number of rejected packages exceeds the acceptance number for the appropriate sampling plan of Standard MIL–STD–105, the lot represented by the sample shall be rejected.

4.2.2 *Examination of brick.* Each of the sample bricks selected in accordance with 4.1.4 shall be visually and dimensionally examined to verify compliance with the requirements of this specification. Any brick in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective bricks in any sample exceeds the acceptance number in that sample, the lot represented by the sample shall be rejected.

4.2.3 *Rejected lots.* Rejected lots may be resubmitted for Government inspection provided the contractor has removed or repaired all nonconforming packages or bricks.

4.2.4 *Lot acceptance tests.*

4.2.4.1 *Place of test.* Unless otherwise specified by the bureau or agency concerned, lot acceptance tests shall be conducted at the place of manufacture, by the manufacturer in the presence and under the supervision of the Government inspector. Manufacturers shall provide the necessary laboratory facilities satisfactory to the inspector, or shall engage the services of a commercial testing laboratory acceptable to the inspector.

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4.2.4.2 *Action in case of failure.* A rejected lot may be resubmitted for Government inspection only after the manufacturer, after being informed of the reasons for rejection, has so reworked the entire lot as to remove or correct all nonconforming material.

4.2.4.3 *Density.* Each of 10 of the sample bricks selected in accordance with 4.1.5 shall be tested for density. If the average of all 10 samples is more than that specified in 3.3 the lot shall be rejected.

4.2.4.4 *Pyrometric cone equivalent.* Two test cones shall be made from a composite sample from portions of five sample bricks selected in accordance with 4.1.5. If either one of the test cones fails to conform to 3.4, a retest shall be made on two cones made from portions of five additional bricks. If any retest cone fails, the lot shall be rejected.

4.2.4.5 *Modulus of rupture.* Each of 10 sample bricks selected in accordance with 4.1.5 shall be tested for modulus of rupture. If two or more of the samples have values less than 45 pounds or if the average of all 10 samples is less than 65 pounds the lot shall be rejected.

4.2.4.6 *Small lots.* The tests specified in 4.2.4.3, 4.2.4.4 and 4.2.4.5 shall be omitted when the lot consists of less than 1,000 bricks.

4.2.5 *Tests at a Government laboratory.* The samples selected in accordance with 4.1.6 shall be subjected to the tests specified in 4.3.

4.2.5.1 *Place of tests.* Unless otherwise specified by the bureau or agency concerned, tests shall be conducted at the Naval Boiler and Turbine Laboratory, Naval Base Station, Philadelphia 12, Pa.

4.2.5.2 *Action in case of failure.* Except as herein specified, acceptance and rejection of lots shall normally be on the basis of sampling, examination and tests of 4.1.4, 4.1.5, 4.2.1, 4.2.2 and 4.2.4, and acceptance shall not be withheld pending receipt of test reports on test samples. However, upon receipt of an unsatisfactory test report on a test sample, the Government inspector shall select additional samples as specified in 4.1.6 from every subsequent lot offered for delivery. The samples so selected shall be submitted to the Government laboratory, and shall there be subjected to the test or tests wherein failure was observed. Lots shall then be accepted only upon receipt of a satisfactory test report on the samples so selected. This additional testing shall be discontinued and lot acceptance returned to the normal basis when four successive lots have been accepted.

4.3 Test procedures.

4.3.1 *Density.* Bricks shall be dried to constant weight at a temperature of 220 to 230°F. and then weighed individualy to the nearest 0.01 pound. The volume of each brick shall be computed from the average dimensional data. From these data the density in pounds per 9 inch equivalent shall be computed.

4.3.2 *Pyrometric cone equivalent.* Determination of pyrometric cone equivalent (PCE) shall be made using standardized procedure.

4.3.3 *Modulus of rupture.* The single upper knife edge of the modulus of rupture machine shall be brought to bear on the center of one 9 by 4½ inch face of the brick. The lower knife edges shal be spaced to have a span of 7 inches. The knife edges shall have a radius of ⅝ inch. The modulus of rupture shall be calculated from the following formula:

$$R = \frac{3wl}{2bd^2}$$

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Where:

R = the modulus of rupture, in p.s.i.

w = the total load, in pounds, at which the specimen fails.

l = the distance between supports, in inches.

b = the width of specimen, in inches.

d = the depth of specimen, in inches.

4.3.4 *Thermal conductivity.* Determination of thermal conductivity (k) shall be made using standardized procedure.

4.3.5 *Simulative service test.* The general arrangement of the simulative service test furnace is shown on figure 1. Twelve test bricks, including four bricks measured to the nearest 1/32 inch at four points for each dimension as shown on figure 2 and averaged to determine their dimensions, shall be laid up without mortar as back-up insulation for a 2½ inch thick dense, 9-inch straight fire-brick panel as shown on figure 3. The test bricks shall be backed up with 1 inch of high-temperature insulating block and a steel plate for support. The furnace shall be heated so as to maintain a temperature on the hot face of the test bricks at 2,500°F. for 24 hours. At the conclusion of this test the panel shall be allowed to cool to room temperature, examined for evidence of fusion, warping, and shrinkage, and then photographed.

4.3.6 *Linear change.* After completion of the simulative service test specified in 4.3.5, each of the four bricks shall be measured at three points on the 9 inch dimension of the hot face to the nearest 1/32 inch. These average data, together with the respective average data before testing, shall be used in determining the linear change.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 *Level A.* Not more than 25 straight 9 by 4½ by 2½ inch brick or equivalent number of split bricks or special shapes shall be packed in wood-cleated plywood, nailed wood or fiberboard boxes conforming to Specification PPP–B–601 (overseas type), PPP–B–621 or JAN–P–108 compliance symbol V3c or V3s, respectively. Void spaces within the container shall be filled with cushioning material to prohibit movement within. Wood and wood-cleated type boxes shall be lined with sealed case liners conforming to Specification MIL–L–10547. Fiberboard boxes shall be closed and waterproof sealed with tape as specified in the appendix to Specification JAN–P–108. The boxes shall be strapped as specified in the appendix of the applicable box specification.

5.1.2 *Level B.* Bricks, in quantities specified for level A, shall be packed in suitable style corrugated or solid fiberboard boxes conforming to class 2 of Specification PPP–B–636. The boxes shall be strapped with one girthwise and one lengthwise tension-tied steel band of a minimum ⅜ by 0.015 inch steel strapping conforming to Specification QQ–S–781. Void spaces within the containers shall be filled with cushioning material to prohibit movement within.

5.1.3 *Level C.* The bricks shall be packed in a manner to insure carrier acceptance and safe delivery to destination at the lowest applicable rate. Containers shall conform to the Uniform Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

5.2 Marking. In addition to the following and any special marking required by the contract or order, shipping containers shall be marked in accordance with Standard MIL–STD–129:

KEEP IN DRY PLACE

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

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(a) Title, number, and date of this specification.

(b) Dimensions required (see 3.2).

(c) Level of packing required (see 5.1).

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodians:

Army—Corp of Engineers
Navy—Bureau of Ships

Other interest:

Navy—Y

Preparing activity:

Navy—Bureau of Ships (Project 705-156)

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General arrangement of furnace for simulative service test (insulation brick).

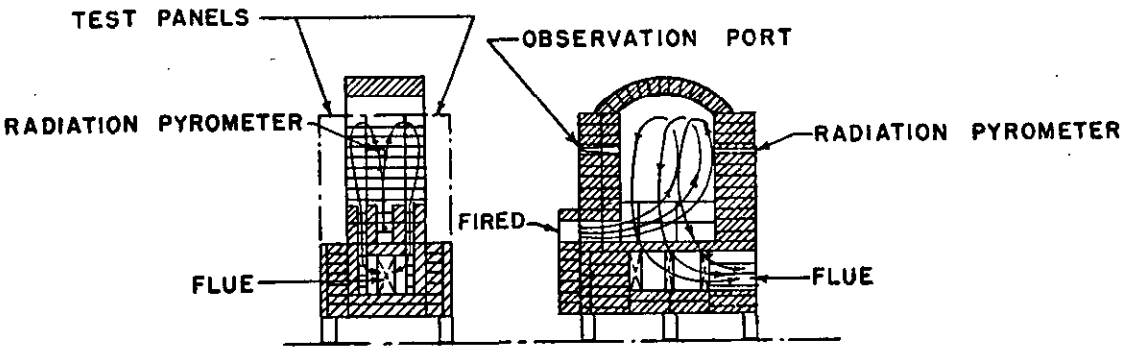


FIGURE 1. *Refractory testing furnace.*

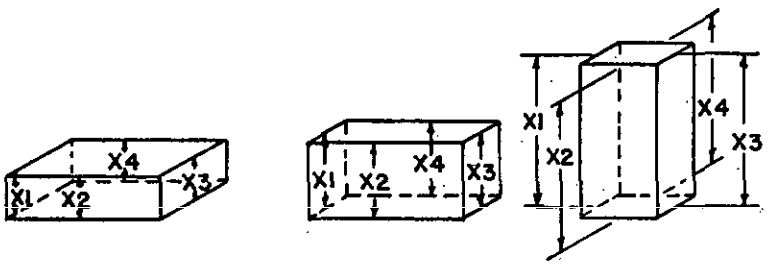


FIGURE 2. *Method of test brick measurement.*

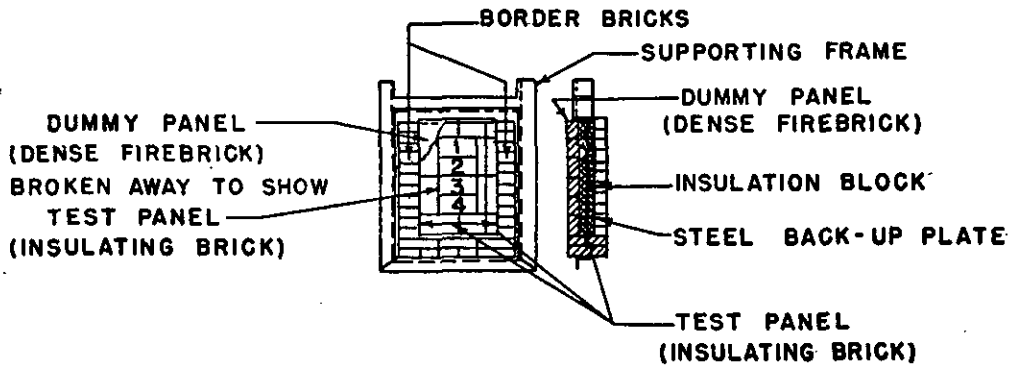


FIGURE 3. *Test panel.*

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